(14) Find the matrix A which satisfies the matrix relation $A\begin{bmatrix}1 & 2 & 3\\ 4 & 5 & 6\end{bmatrix}=\begin{bmatrix}-7 & -8 & -9\\ 2 & 4 & 6\end{bmatrix}$.

(15) If $A^T=\begin{bmatrix}4 & 5\\ -1 & 0\\ 2 & 3\end{bmatrix}$ and $B=\begin{bmatrix}2 & -1 & 1\\ 7 & 5 & -2\end{bmatrix}$, verify the following

(i) $(A+B)^T=A^T+B^T=B^T+A^T$ (ii) $(A-B)^T=A^T-B^T$ (iii) $(B^T)^T=B$.

(16) If A is a 3×4 matrix and B is a matrix such that both A^TB and BA^T are defined, what is the order of the matrix B?

(17) Express the following matrices as the sum of a symmetric matrix and a skew-symmetric matrix:

(i) $\begin{bmatrix}4 & -2\\ 3 & -5\end{bmatrix}$ and (ii) $\begin{bmatrix}3 & 3 & -1\\ -2 & -2 & 1\\ -4 & -5 & 2\end{bmatrix}$.

(18) Find the matrix A such that $\begin{bmatrix}2 & -1\\ 1 & 0\\ -3 & 4\end{bmatrix}A^T=\begin{bmatrix}-1 & -8 & -10\\ 1 & 2 & -5\\ 9 & 22 & 15\end{bmatrix}$.

(19) If $A=\begin{bmatrix}1 & 2 & 2\\ 2 & 1 & -2\\ x & 2 & y\end{bmatrix}$ is a matrix such that $AA^T=9I$, find the values of x and y.

(20) (i) For what value of x, the matrix $A=\begin{bmatrix}0 & 1 & -2\\ -1 & 0 & x^3\\ 2 & -3 & 0\end{bmatrix}$ is skew-symmetric.

(ii) If $\begin{bmatrix}0 & p & 3\\ 2 & q^2 & -1\\ r & 1 & 0\end{bmatrix}$ is skew-symmetric, find the values of p, q, and r.

(21) Construct the matrix $A=[a_{ij}]_{3\times 3}$, where $a_{ij}=i-j$. State whether A is symmetric or skew-symmetric.

(22) Let A and B be two symmetric matrices. Prove that $AB = BA$ if and only if AB is a symmetric matrix.

(23) If A and B are symmetric matrices of same order, prove that

(i) $AB + BA$ is a symmetric matrix.

(ii) $AB - BA$ is a skew-symmetric matrix.

(24) A shopkeeper in a Nuts and Spices shop makes gift packs of cashew nuts, raisins and almonds.

Pack I contains 100 gm of cashew nuts, 100 gm of raisins and 50 gm of almonds.

Pack-II contains 200 gm of cashew nuts, 100 gm of raisins and 100 gm of almonds.

Pack-III contains 250 gm of cashew nuts, 250 gm of raisins and 150 gm of almonds.

The cost of 50 gm of cashew nuts is ₹ 50, 50 gm of raisins is ₹10, and 50 gm of almonds is ₹ 60. What is the cost of each gift pack?

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